

Making Security Manifest

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Abstract

Information is necessary for a functioning market. Yet currently, security information is held by specialists, and no simple means exist to transfer that expertise to the average user. Secure systems are luxury goods, available only to those with the most discerning clientele capable of distinguishing, or for those businesses where the high cost of failure justifies large security investments (Dean 2003). In this paper we describe a mechanism to make it possible for a consumer to easily view the security of a remote system or their own.

Enabling ubiquitous, but not intrusive, security reporting can allow the consumer market for security to function. We refer to this as “manifest security.” We propose three layers for the creation of a security market.

- First there is the specification of a XML scheme to allow for extensible expressions of trust as these develop on-line over time.
- Second, there is the reputation or recommending layer. This requires the capacity to use collaborative filtering or simple rating/reputation systems. This consists of the participants and selection of participants.
- Third, there is the reporting layer. In the first implementation, the reporting layer would be a browser applet, or a stand alone display.

Sufficient market pressures or other incentives are necessary to enable the diffusion of this overlay infrastructure. Yet there must be a proposal and proof of concept to enable adoption.

Introduction & Motivation

That security is a market failure is the subject of some agreement. A commonly proposed solution to rectify this market failure is the establishment of a liability regime for computer security (Anderson 94). Other potential remedies for market failures in security include creating tradable permits (Camp & Wolfram 2000) and insurance markets for business risks (e.g. Blakely 2002). Most of the discussion, however, centers on the enterprise and organizational actors, the traditional consumer or high-end security goods. Yet security is also a key factor in consumer behavior online, remaining a chief concern for online consumers (Deringer 2002). Information has the potential to transform security from a rare luxury good to a common commodity.

Reputation markets have a result in terms of price. While it is intuitively obvious that reliable merchants can charge more than unreliable merchants; it is also supported by a game theoretic approach to bidding in the face of reputation information (Dellarocas 2003), as well as empirical evidence from online auctions (Resnick & Zeckhauser 2002). We assert that security is another facet of reliability in the larger context of electronic interactions, exposing the actors to failure of not only the transaction, but potentially their entire system and loss of valuable information.

The majority of security literature focuses on the administrator level, preventing attacks on the technical side with a level of expertise outside the range of most non-technically trained users. If

security is salient to their user experience, some accessible user-friendly mechanism should allow users to accept some responsibility for the security of their online interactions.

Descriptive Mechanism

Obviously, the development of a flexible, extensible XML lexicon describing a full range of security issues is a non-trivial task. Development should involve a full range of interested parties, and adhere to a fully open, participatory process. Where possible, the scheme should reflect easily verifiable information that can cleanly be stated to be either true or false (i.e. is the latest patch installed). Grammatical structures should reflect the ever-evolving nature of security requirements and be as open as possible. The development process of W3C's Platform for Privacy Preferences can be held as a model for cooperative and effective descriptive language design.

Reputation and Recommender Systems

If individual users lack the expertise and resources to assess the security of the systems with which they interact, then the next best solution is to rely on trusted experts. Of course, it is never so simple: one seldom knows whether to trust an expert. Information economics has recently seen a rich development in the field of recommender or reputation systems as a means of determining trustworthiness. Recommender systems make up the backbone of product recommendation mechanisms (i.e. Amazon); they also serve as crucial social norms enforcers in public forums (i.e. Slashdot). At its heart, such a system consists of some means of allowing some subset of a community to voice their opinions on the value of the agent or product being evaluated; this feedback is aggregated and used by other actors in their relationship to the object of the feedback. Such mechanisms can create records for developing trust in a society of strangers (Dingledine 1999; Camp 2003).

We can imagine expertise varying across a range of dimensions, each defined in terms of a security concern of a user. For instance, a reader of an online newspaper may want to know that the site has not been defaced; a user of chat rooms may want to know that her anonymity is being protected; a consumer of health information would care about the security of personal information collected. Someone shopping on the ACLU website might care about all three. Along each of these dimensions, a security analysis of a given system can be offered by any party. Other actors can support or take exception to this evaluation, and the reputation system can aggregate these sentiments.

Each user may choose to select a reputation provider. The obvious candidates for reputation systems include incident response teams, software vendors, and security consultants. Incident response teams are interested in increasing network security. Software vendors can offer high ratings to companies that download and install patches in a timely manner. Security consultants can offer their expertise to a broader market.

While in the privacy field, third party verification in the form of Privacy Seals has met with limited success, we believe that by balancing third party reputations with the feedback of a recommender system, a useful measure of aggregated trust can be established in a security evaluation system. Recommendations are not entirely unproblematic, however, as perverse incentives may tempt certain actors to downgrade the security ratings of their competitors.

Furthermore, unlike a book or an auction seller, refutation of a positive recommendation (or, to a lesser extent, a negative comment) will not be immediately confirmed for feedback into the system. Similarly, unlike a purchase there may be no direct cost for a particular entity to create a recommendation.

While widespread participation creates its own set of risks, having a large number of users in addition to a small set of experts is optimal. Increasing the number of observers increases the probability that an action is observed. For example, many viewers increase the chance of viewing a webpage defacement, receiving notification of a need for a new credit card, or experiencing fraud directly connected to a subverted server. However, the risk of a security failure of any one system is relatively low; we thus cannot depend solely on a self-correcting reputation system for security analysis. Trusted third parties can claim the authority to evaluate the expertise of evaluators, inspiring trust, or assume the role themselves.

The system may collect ratings with simple UDP requests that are sent in parallel with http requests. Alternatively the user could have a learning agent that trusts its own user assertion of positive and negative on-line experiences and collects information of others. In this case, reputations for frequently visited pages would be updated by the agent, and the agent could build from the person's browsing history to check on the sites regularly visited. Each system has its own benefits and drawbacks in inspiring trust. Optimally, some combination of external expert recommendations, reputation systems and logged personal preferences would enable a user to feel confident about the security reputation of the system with which she is interacting. A non-security analog would be a reader who uses the New York Times Book Review, epinions.com book rating system and amazon.com's personal recommendations page to select her next volume.

Displaying the Result

Once a level of security has been determined, the user needs a relatively simple and real-time interface. We have no reason to believe that users will undertake this research themselves, and thus claim that an unobtrusive interactive agent is needed to deliver security information for effective use. A user will need to interact with the security system in two ways.

First, she will need to configure the system to reflect her preferences in terms of security and the parties she trusts to evaluate the security of the internet hosts with which she will be interacting.

Second, she will need to learn whether or not a specific action conforms to these preferences. Three possible models present themselves for feedback mechanisms. First, the user can be alerted by a simple Boolean trust-worthy / not-trustworthy notification. Since security is hardly such a one-dimensional variable, the user should have some idea of what being trust-worthy entails, either by understanding what her trusted agent is examining, or having expressed her security preferences to an agent that interprets the security evaluation from the reputation mechanism. Second, the multi-dimensional aspects of security can be converted into a one-dimensional scale, expressed as a number in a given range or a color in a given spectrum. This also implies some awareness of what more secure or less secure means to the user, but allows gravitation towards a greater degree of security. Finally, a multi-dimensional feedback mechanism, perhaps in the presence or absence of small icons, can alert the user to various levels

of security in terms of, for example, encryption and back-end database protection. After a while, a user will learn to submit her own security recommendations or experiences as a rating into the reputation mechanism.

Two models we use as inspiration are AT&T's Privacy Bird using W3C's P3P standard, and the stand-alone NVisionIP network security tools. The Privacy Bird, a browser plug-in that employs visual and auditory cues to keep the user informed about the privacy promised by a viewed web page (see <http://www.privacybird.com>). The Privacy Bird allows users to express their privacy preferences in easy to understand language, and also provides even simpler abstractions of 'low,' 'medium,' and 'high' levels of privacy. It can be positioned around the screen and its visual and noise cues can be set for maximum convenience and minimum annoyance. The NVisionIP security tools, developed at NCSA, allow users to easily interpret a vast amount of network traffic administration with visual pattern and color information (Lakkaraju & Yurcik 2003).

The two major critiques of the Privacy Bird in a recent study are a lack of P3P implementation across the web and an inability to trust the privacy policies available (Cranor et al 2002). The latter is addressed in a manifest security model by a transfer of trust to a consumer-selected trusted entity; the issue of implementation will be addressed below. The key features desired in this system of accessibility and multiple levels of customizability offer easy-to-use security information and allow for more responsible and informed internet use.

Potential Market Configurations

This paper offers three models under which manifest security might be propagated across a large segment of the internet-using non-expert population: fear of attack, de facto or de jure regulation and the more market-centered liability expansion.

As computer security becomes a more mainstream topic in the media, and computer security threats rise, it is possible that a tipping point could be reached in terms of mass perception of security exigencies. Several high-profile security debacles, for example, might make it easier to conceive of a personal security threat for the average user. In this case, public demand should drive a market solution to some security issues. Users would actively seek reporting clients, and the services of commercial or non-profit security evaluation agencies, if they were available. Manifest security offers a possible solution that is less extreme than a wholesale shift to secure systems such as TCPA or Palladium, and allow greater user autonomy.

Alternatively, browsers could start being shipped enabled with some manifest security technology built in and turned on. This could come from a government mandate for some national security policy or, as was the case with P3P, acceptance of the user-end technology by the market leader in browsers. Without a sufficient demand, however, it will be much more difficult to build the reputation infrastructure, and the technology might languish as a not-wholly assimilated system. However, the largest browser provider may have an incentive to provide ratings that underestimate the security of competitor's products.

Finally, we can speculate about an environment in which the liability standards that have been advocated for quite some time are indeed adopted. If parties responsible for security-related damages are held accountable, it is likely that some of this accountability will filter down to the

end user. Machines hooked up to always-on, high-speed internet access, for example, have recently been implicated in the rise of Distributed Denial of Service attacks. If the victims of these attacks hold the ISP's liable for damages, ISPs would thus have a very strong incentive to encourage proper security practices in their customers. They would encourage both the use of interactive agents and the acceptance and reputability of trusted security experts, strengthening both sides of the system. We can imagine ISP or insurance firm support of trusted third party reviewers, and even the development of manifest security tools as web services at the ISP gateway level.

Each of these scenarios can be analyzed to speculate what a large-scale implementation of manifest security might look like. This research is currently in progress.

Discussion and Benefits

Under this guise, security is taken as a super-set of currently applied observable values: is the latest patch installed, did the latest audit find leaking data or is this site using an easily breakable key. (For example, a survey of keys in 2001 found keys as short as 40 bits, see van Somerson 2001). Manifest security is the means by which a user of information systems can be sure that information being sent and received is valid and can be trusted. Market forces develop when demand awareness are high enough to place consumer value on security information; that information could be paid for by the consumer seeking security, or a remote host seeking to accrue a reputation in order to earn the trust of customers.

A recent study of P3P users found that they were thinking more about privacy (Cranor et al, 2002). Similarly, more responsible, informed views for better decision-making are needed for information security. Making security manifest to the user will generate more public awareness of security, increase information, and thereby enable the security market to function in a way as to serve consumer interest.

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